Polynomial Voxels on an Octree

Blocks between Unequal Cells, Refinement by the Two-Term Error Estimate, and the Basis for a Medium that is Piecewise Constant on the Leaves

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Abstract

This document archives the octree material removed from the paper on the error of polynomial voxels [1] when that paper was narrowed to the error analysis on uniform grids. It is kept as a record, not as a paper. It holds the octree itself (leaves of different sizes, the exact blocks between them as sums of blocks between equal cells, and their reduction by the symmetry of the cube), the two-term error law measured on trees, the refinement rule built on the two terms and its test on a sphere with a localised feature, the test on a medium that is piecewise constant on the leaves, and a summary of the two-scale Galerkin refinement studied afterwards. Why the line was set aside is stated first (§1). The error law that the refinement uses is that of Nestor [1], whose equations and sections are cited here in words.

1 Why the octree was set aside

- *Size.* The tree solver is dense and stops near 20 000 unknowns: about 2200 constant leaves or 550 degree-one leaves. Every result below is on a small body. A tree of 344 leaves of four sizes is assembled and solved in about three minutes; an equal-cell block costs about 50 ms.
- *A fast solver is a project of its own.* Its parts are published: the fast multipole method for volume potentials on an octree [2], the elastic kernels in fast multipole methods for boundary integral equations [3, 4], and, for cells mapped to a uniform grid, the adaptive integral method [5]. None is implemented here.
- *The refinement did not improve further.* The two-scale Galerkin indicator of §6 builds the trees of the two-term rule and reaches the same accuracy, at the cost of a solve per step where the two-term rule needs none.
- *The largest gain is self-consistent.* The factor of 65 of §5 is against a reference computed by the same scheme with degree-two fields, on a body whose field is not smooth near the edges and corners of its cubes.

- *Uniform grids do well.* Uniform grids of Legendre cells, applied by fast Fourier transforms, reach high accuracy on a graded sphere without a tree [6]; the tree paid only where the variation of the medium filled a few per cent of the volume.

2 The octree

Leaves. The body is covered by cubes, the leaves of an octree: leaf ℓ has centre $\mathbf{x}_\ell$ and half-width h_ℓ (side $d_\ell = 2h_\ell$), and the half-widths are in ratios that are powers of two. With $\boldsymbol{\xi} = (\mathbf{x} - \mathbf{x}_\ell)/h_\ell$ the local coordinate, the leaf carries the nine-component state (displacement and strain) in the Legendre products $Q_a(\boldsymbol{\xi}) = P_{a_1}(\xi_1)P_{a_2}(\xi_2)P_{a_3}(\xi_3)$ of total degree $|a| = a_1 + a_2 + a_3 \le p_\ell$, which is 1, 4 or 10 functions for $p_\ell = 0$, 1 or 2, and the contrast in the same products to degree r_ℓ. The two degrees may differ from leaf to leaf. A leaf with $p = r = 0$ is a Haar cell: a constant medium and a constant field. We call it a constant cell, a leaf with $p = r = 1$ a degree-one cell (the constant and the three linear polynomials, ξ_1, ξ_2, ξ_3), and a leaf with $p = r = 2$ a degree-two cell (those four and the six quadratic products, $P_2(\xi_i)$ and $\xi_i\xi_j$ with $i < j$; ten in all).

The system. Tested with the same functions, the volume integral equation becomes

$$M_\ell\,\psi_\ell - \sum_m \mathbf{K}(\ell, m)\,E_m\,\psi_m = \langle Q, \psi^0\rangle_\ell, \tag{1}$$

with ψ_ℓ the coefficients of the state in leaf ℓ, M_ℓ the Gram matrix of its functions, E_m the projected contrast of leaf m acting on its field, ψ^0 the incident wave and $\mathbf{K}(\ell, m)$ the block that couples the two leaves: the Green's tensor integrated against the functions of both [6].

Blocks between leaves of different sizes. A polynomial on a cube is a polynomial of the same degree on each of the cube's descendants, so the block between a large leaf and a small one is an exact, finite sum of blocks between equal cells: the large leaf is replaced by its descendants at the size of the small one, with its polynomials re-expanded (Figure 1c). No integral is needed beyond those between equal cells. This is the two-scale relation of the multiwavelet basis [7]. Against six-dimensional quadrature these blocks agree to 10^{-9} for leaves one and two levels apart, and on a tree whose leaves are all equal the solver reproduces the uniform solver to 10^{-11}.

Symmetry. Two pairs of leaves whose relative positions are images of one another under a reflection or a permutation of the axes have blocks related by the representation of that operation on the polynomials and on the nine components, whatever the two sizes, because the operation acts on both cubes alike. One block is therefore computed for each orbit of the cube's group of 48 operations. On a tree of 344 leaves of three sizes this reduced the assembly from 125 s to 65 s with an unchanged solution, and made trees with four sizes practicable.

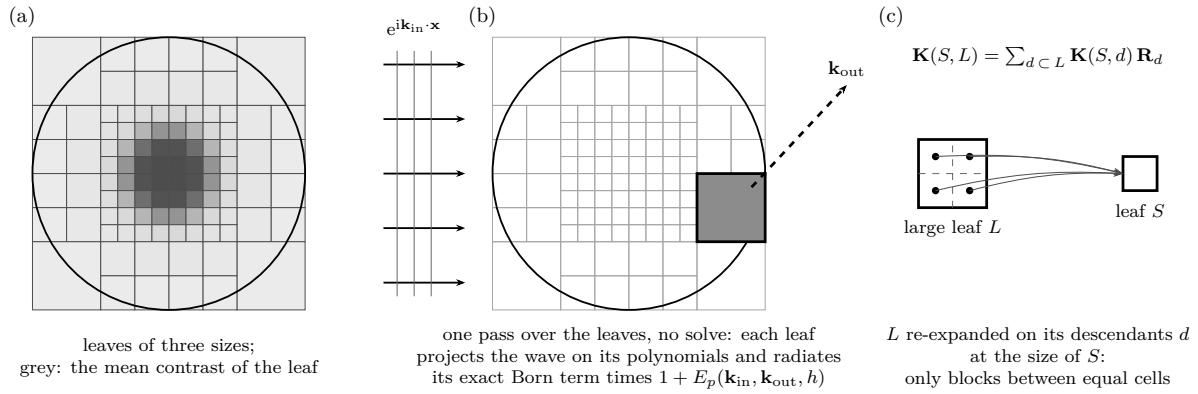


Figure 1. The octree, drawn as its two-dimensional slice. (a) The leaves on a body with a weak halo and a compact graded feature, refined where the feature's contrast varies, in three sizes; the grey level is the leaf's mean contrast, which the leaf holds as a polynomial of degree r. (b) The term of first order needs no solve: each leaf holds the projection of the incident wave onto its own polynomials and radiates its exact Born field times the wave factor $1 + E_p$ of Nestor [1], which depends only on the leaf's half-width and the two wavevectors, so the whole term is one pass over the leaves. (c) The block between a large leaf L and a small leaf S is the sum of the blocks between S and the descendants d of L at the size of S, with the polynomials of L re-expanded on them ($\mathbf{R}_d$): only blocks between equal cells are ever integrated. The tree is generated by `figures/make_fig_octree.py`.

3 The two-term law on trees

The law of Nestor [1], that the error of the far field less its first-order error is the nonlinear fraction of the response times the relative projection error $\mathscr{E}_r$ of the medium, times a factor of order one, was measured on uniform grids and on trees alike. The full table is kept here; the error paper keeps its uniform rows. The test body is a sphere of radius a with a uniform core of radius $0.75a$ and a shell across which the contrast falls smoothly to zero, at $k_S a = 0.5$; the error is the largest difference of the far field from the exact one over nine scattering angles, relative to its peak. For constant cells the ratio of error to projection error varies by a factor of 1.93 across the grids; the rest over the projection error varies by 1.06, and it is 0.0151 to 0.0152 on all four trees. For degree-one cells the first-order error is small and both ratios are near 0.0146. The constant is close to the nonlinear fraction of the exact response, $\nu = |u - T_1|/|u| = 0.0151$, on trees as on uniform grids.

4 Refinement by the two terms

The rule. Each leaf has two indicators, both computed before any solve. The *medium* indicator is the nonlinear fraction ν times the leaf's share of the projection error, $\nu \int_\ell (f - \Pi_r f)^2 \,\mathrm{d}V / \int f^2 \,\mathrm{d}V$. The *wave* indicator is the leaf's own first-order error, computed without a solve [1], relative to the peak of the Born far field. A leaf is split into its eight children while the sum of the two exceeds a tolerance. The nonlinear fraction is estimated on a coarse tree that resolves the medium, from two products with its matrix and no solve [1]. The experiments below were first run with the earlier estimate, from one solve on that tree; the trees and their errors are the same with either, since ν enters the rule only through the medium indicator and the two estimates differ by up to

Table 1. The error of the far field in two terms, on the sphere with a graded shell. The rest is the error less the first-order error, which is computed without a solve. $\mathscr{E}$ is the relative projection error of the medium at the degree of the cells [1]. The nonlinear fraction of the exact response is 0.0151.

cells	grid	leaves	error	error / $\mathscr{E}$	rest / $\mathscr{E}$
constant ($p = r = 0$)	uniform	64	8.17×10^{-3}	0.0309	0.0149
	uniform	184	3.18×10^{-3}	0.0235	0.0151
	uniform	408	1.61×10^{-3}	0.0196	0.0154
	uniform	1256	6.77×10^{-4}	0.0166	0.0158
	tree	232	3.29×10^{-3}	0.0321	0.0152
	tree	856	9.28×10^{-4}	0.0210	0.0152
	tree	1360	6.03×10^{-4}	0.0210	0.0151
	tree	1696	5.75×10^{-4}	0.0222	0.0151
degree one ($p = r = 1$)	uniform	64	8.99×10^{-4}	0.0147	0.0148
	uniform	184	4.62×10^{-4}	0.0146	0.0146
	uniform	408	2.20×10^{-4}	0.0146	0.0146
	tree	184	5.16×10^{-4}	0.0142	0.0148
	tree	352	2.20×10^{-4}	0.0146	0.0145

a fifth, and only the predicted errors change. The predicted error of a tree is the first-order error of the whole tree, summed over the leaves as complex amplitudes, plus $\nu \mathscr{E}_r$. It uses no reference solution.

A localised feature. The test body is a sphere of radius a whose contrast is a weak halo, falling smoothly from 5% of the central value to zero across the whole radius, plus a compact feature in which it falls from the central value to the halo between $0.1a$ and $0.3a$: under 3% of the volume. Its exact solution is known. At $k_S a = 1$, with leaves no smaller than $a/32$, the errors are those of Table 2. The rule that uses the medium alone refines a leaf while its share of the projection error exceeds a tolerance.

Constant cells. The rule that uses the medium alone fails: with 512 leaves its error is twice that of the uniform grid of 408 cells. It refines the feature and leaves the halo coarse, and the halo is where the first-order error is, in leaves whose medium is too smooth for the rule to see. The rule with two terms sees it. Its tree of 260 leaves has the error of the uniform grid of 408, and against the trend of the uniform grids its trees have the same error with about 35% fewer cells at the coarse end and an error 15 to 20% lower at the fine end. The saving is modest, and the reason is the cell: a constant field has a second-order error in every leaf, so the halo needs cells however smooth its medium is.

Degree-one cells. Here the tree pays. The tree of 176 leaves has eleven times less error than the uniform grid of 184 cells. By the trend of the two uniform grids its error would need about 730 uniform cells, four times as many, and that of the tree of 452 leaves about 1130; these uniform equivalents are extrapolated, not computed. The fourth-order error of a degree-one field lets the halo stay coarse, and the tree then follows the medium. The two-term rule improves on the medium alone by about 15% at about 340 leaves.

Table 2. The localised feature: leaves, the error predicted without a reference, and the error against the exact solution. The nonlinear fraction estimated on a coarse tree, with no solve, is 0.0059 for constant and for degree-one cells, the exact value; the earlier estimate from a solve was 0.0051 for constant cells.

cells	grid	leaves	predicted	error
constant	uniform	408	1.15×10^{-2}	7.15×10^{-3}
	uniform	1256	4.50×10^{-3}	2.92×10^{-3}
	medium alone	344	1.61×10^{-2}	1.55×10^{-2}
	medium alone	512	1.57×10^{-2}	1.53×10^{-2}
	medium alone	736	5.96×10^{-3}	5.49×10^{-3}
	two terms	260	8.31×10^{-3}	7.18×10^{-3}
	two terms	372	6.93×10^{-3}	5.38×10^{-3}
	two terms	752	4.53×10^{-3}	3.78×10^{-3}
	two terms	1780	2.22×10^{-3}	1.75×10^{-3}
degree one	uniform	184	2.86×10^{-3}	2.73×10^{-3}
	uniform	408	8.36×10^{-4}	6.77×10^{-4}
	medium alone	176	2.44×10^{-4}	2.47×10^{-4}
	medium alone	344	1.95×10^{-4}	1.96×10^{-4}
	two terms	260	1.73×10^{-4}	1.68×10^{-4}
	two terms	336	1.57×10^{-4}	1.66×10^{-4}
	two terms	452	1.09×10^{-4}	1.14×10^{-4}

The prediction. The predicted error is above the true error on all eleven grids of constant cells computed for this body (nine of them in Table 2), by 3% to 61%. For degree-one cells the ratio of predicted to true is 0.95 to 1.23 on nine grids: close, but not a bound.

5 A medium that is piecewise constant on the leaves

The pairing of a constant medium with a polynomial field belongs to the medium that is constant in each leaf. The test body is a cube of half-side B with a uniform contrast a quarter of the central value, holding at its centre a cube of half-side $B/4$ with the full contrast, at $k_S B = 2$. A tree of 120 leaves represents it exactly with one constant per leaf ($\mathscr{E}_0 = 0$): 56 leaves of half-width $B/4$ and 64 of half-width $B/8$. The uniform grid that does the same has 512 cells. No exact solution is known for this body; the reference is the tree with degree-two fields, whose own first-order error is predicted to be 2×10^{-7} (Table 3). With the medium exact there is no second term, and the first-order prediction is within 13% of the measured error in all seven cases and within 4% in five. The tree with a degree-one field has an error of 8.4×10^{-5} with 4320 unknowns; the uniform grid of constant cells has 5.5×10^{-3} with 4608: 65 times more. Raising the field to degree one in the large leaves alone lowers the tree's error from 1.5×10^{-2} to 3.2×10^{-3}: a large leaf needs a richer basis for the wavefield, not for the medium. And where the faces of the inner cube fall inside cells, the degree-one field is worse than the constant one, as on the smooth body; for the scalar problem Anderson et al. [8] likewise find that high order is lost at jumps of the medium unless the grid follows them.

Table 3. A cube within a cube, the contrast constant in every leaf. The first-order error is predicted without a solve; the error is against the tree with degree-two fields. In the last two rows the faces of the inner cube fall inside the cells of the grid.

grid and field	cells	unknowns	predicted	error
tree, constant field	120	1080	1.54×10^{-2}	1.53×10^{-2}
tree, degree one in the 56 large leaves	120	2592	3.24×10^{-3}	3.21×10^{-3}
tree, degree-one field	120	4320	8.07×10^{-5}	8.42×10^{-5}
uniform, constant field	512	4608	5.54×10^{-3}	5.51×10^{-3}
uniform, degree-one field	512	18432	8.17×10^{-6}	9.25×10^{-6}
uniform, constant field	64	576	1.81×10^{-2}	1.79×10^{-2}
uniform, degree-one field	64	2304	3.01×10^{-2}	2.92×10^{-2}

The choice. The two cases give the rule. Where the medium is piecewise constant and the tree can follow its interfaces, hold it constant in each leaf and spend the degree on the field. Where it varies smoothly, give the medium and the field the same degree and let the tree follow the medium. In both the error is predicted before the solve.

6 The two-scale Galerkin refinement

After the two-term rule, a refinement indicator was derived from the two nested Galerkin problems of a tree and of its children (`docs/2026-10-09-two-scale-galerkin.md`). The residual of the prolonged coarse solution splits exactly into a part driven by the detail of the medium and a part driven by the multiwavelet detail of the field, and the error of the far field is a sum of their shares, cell by cell, weighted by the adjoint solution. On a graded sphere the identities hold to 10^{-13}, the two-scale difference measures the coarse error to 7%, as fourth-order convergence predicts, and a two-level estimate, with one coarse solve and one fine product, gives the error to 0.03 to 0.14% (`scripts/derive_two_scale_galerkin.py`).

On an octree (degree-one cells, $k_S a = 1$, leaves no smaller than $a/32$, the fine operator applied by fast Fourier transform on the finest uniform grid; `scripts/pilot_octree_two_scale_refinement.py`), the indicator with a tolerance per leaf builds the trees of the two-term rule, and marking by the Dörfler criterion reaches the same band (Table 4). Neither form

Table 4. Refinement on the localised feature of §4, degree-one cells: leaves and error against the exact solution.

leaves	two-scale indicator		two-term	medium alone	uniform
	tolerance per leaf	Dörfler, $\theta = 0.5$			
120–184	120: 6.8×10^{-4}	134: 5.4×10^{-4}		176: 2.47×10^{-4}	184: 2.73×10^{-3}
218–260	260: 1.68×10^{-4}	218: 2.9×10^{-4}	260: 1.68×10^{-4}		
336–344	344: 1.66×10^{-4}	337: 1.74×10^{-4}	336: 1.66×10^{-4}	344: 1.96×10^{-4}	
408–533		533: 1.19×10^{-4}	452: 1.14×10^{-4}		408: 6.8×10^{-4}

improves on the two-term rule on this body. The two-scale difference, which needs no reference, is 73 to 97% of the true error on trees above 100 leaves, but each step needs a coarse solve, a

fine product and a second coarse solve, four to five minutes here, where the two-term rule needs no solve.

7 Status of this material

- *Established against exact solutions, by one implementation.* The law on trees (Table 1) and the refinement (Table 2): one frequency each, two bodies, both spheres with radial profiles.
- *Self-consistent only.* The cube within a cube (Table 3) has no exact solution; its reference is the same scheme with degree-two fields.
- *Not a bound.* The predicted error exceeded the true error for constant cells in every case, but fell 5% below it for degree-one cells.
- *Earlier work.* The adaptive solvers of Ambikasaran et al. [9] and Malhotra and Biros [2] have not been read in full for an estimate that relates their refinement tolerance to the error of the solution.

The octree, its blocks and its solver are `cubic_scattering/graded_voxel/octree.py`, with tests in `cubic_scattering/tests/test_graded_voxel_octree.py`. Table 1 is `scripts/measure_octree_two_term.py`; Table 2 is `scripts/pilot_octree_two_term_refinement.py`; the constant medium with a richer field is `scripts/pilot_octree_hp_refinement.py`; Table 3 is `scripts/pilot_octree_blocky_body.py`; Table 4 is `scripts/pilot_octree_two_scale_refinement.py`. Their results are stored in `LatexPDFs/archive/OctreeRefinementArchive/figures`, with the data of Table 1 in `LatexPDFs/PolynomialVoxelError/figures`, and in `scratch/two_scale`.

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